Marco Mazzocco

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EDUCATION

Politecnico di Milano 2026 – Now
MSc in High Performance Computing Engineering
- **EIT Digital Master School:** full tuition waiver for the Major in Cloud, Networking Infrastructures and HPC.
- **Curricular Projects:** Agentic AI for RL-Based Scaling and Automated Deployment of AI Applications (ongoing).

Sapienza University of Rome – 110/110 cum Laude 2023 – 2026
BSc in Computer and Control Engineering
- **Relevant Coursework:**
  - Data Structures, Algorithms and Complexity (30/30 cum Laude)
  - Mathematical Programming (30/30 cum Laude)
  - Operating Systems and Computer Networks (30/30)
  - Econometrics (30/30)
- **Curricular Projects:** End-to-end web document parsing and LLM-based evaluation pipeline (FastAPI, Crawl4AI, Pydantic, Docker), selected among the top 5 NLP projects (out of 70+). Extended it with AI agents that generate a site-specific parser for each new website.
- **Bachelor's Thesis:** PageRank: Markov Chains and GPU Computing.
  Code and thesis: github.com/mazzocco51/pagerank-thesis
  - Power iteration and Monte Carlo on GPU (CuPy) on graphs up to 62M nodes and 1.47B edges.
  - Up to 37.1× GPU speedup over CPU; custom CUDA kernel halving graph memory (12 to 6 GB).
  - Personalized PageRank contact recommender on a real social network: at least one correct suggestion for 506 of 5,000 users.

I.I.S.S. Patini Liberatore – 100/100 cum Laude 2018 – 2023
High School Diploma in Scientific Studies
- **Awards & Honors:** National Finalist at the **Italian Mathematical Olympiad** (2021-2022); Candidate for the National Honor **"Alfiere del Lavoro"** (2023) for academic excellence.
- **International Experience:** Participated in a short-term Erasmus+ exchange program in Örebro, Sweden (2022).

SKILLS

Programming: Python, C, CUDA, Java, R, SQL
ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, CuPy, SHAP, Prophet
Infra & Tools: Docker / Docker Compose, GitHub Actions (CI/CD), pytest, PostgreSQL, Git
LLM & Agents: LangGraph, LangChain, n8n, multi-agent systems, LLM-as-a-judge evaluation, local inference (Ollama)
Backend & Web: FastAPI, Pydantic, Streamlit, Crawl4AI, Playwright

LANGUAGES

Italian: Native | **English**: Advanced (Cambridge English C1 Advanced - CEFR, 2022; IELTS Academic 7.5, 2026)

PERSONAL PROJECTS

- **Stock Return Classifier:** Random Forest vs XGBoost vs LSTM on 7 years of AAPL, walk-forward validated (Spring 2026).
- **Trading AI Multi-Agent:** LLM agents vote hourly on BTC, ETH, SOL; a risk layer issues paper trades (Summer 2026).
- **WC2026 Edge Engine:** Monte Carlo + Poisson model vs. Polymarket odds on the 2026 World Cup (Summer 2026).
- **Knowledge Graph Extraction:** LLM pipeline building knowledge graphs, with PageRank centrality analysis (Autumn 2026).

CERTIFICATIONS & TRAINING

- **JPMorgan Chase & Co. - Quant Research Virtual Program (2026):** financial modeling and risk estimation in Python.
- **IBM AI Engineering Professional Certificate (2026):** ML with Python, data preparation for GenAI and LLMs.
- **JUMP, Fondazione RUI (2023 – 2026):** ethics, public speaking, global mindset, team building & leadership.

COMPETITIONS & EVENTS

- **28DIGITAL Business Challenge, Tallinn (2026)**
  Business model and 6-week pilot for GreenPow, an AI compute orchestration startup: ∼8% more GPUs under the same power.
- **REPLY AI Agent Challenge (2026)**
  Ranked in top 5% in online hackathon focused on Multi-Agent Systems for fraudulent transaction detection.
- **CODEMOTION Hack Nation (2026)**
  Hackathon focused on developing a multi-agent AI system for real-time online fact-checking.
- **NTT DATA Hackathon "ikigAI" (2026)**
  Development of ML-based solutions for churn prediction, targeted upselling, and recommender systems.
- **REPLY "Student Tech Clash" (2023, 2024, 2025)**
  Participated in Hackathons dedicated to the development of AI-based application prototypes.